Solutions: Problem Set 9

Fuel Cycles & Material Limits

Problem 1: The Value of "Waste" (Breeding Physics)

(a) Once-Through LWR Energy Extraction

- **Given:** Mass $M = 1,000$ kg. Utilization $U = 0.6\%$ (0.006). Energy density of fission $E_f \approx 1$ MWd/g.
- **Fissioned Mass:**
$$M_{fiss} = 1,000 \text{ kg} \times 0.006 = 6 \text{ kg} = 6,000 \text{ g}$$
- **Total Thermal Energy:**
$$E_{th} = 6,000 \text{ g} \times 1 \text{ MWd/g} = 6,000 \text{ MWd} = \mathbf{6} \text{ GWd}$$

(b) Fast Breeder Reactor (FBR) Energy Extraction

- **Given:** Utilization $U = 60\%$ (0.60).
- **Fissioned Mass:**
$$M_{fiss} = 1,000 \text{ kg} \times 0.60 = 600 \text{ kg} = 600,000 \text{ g}$$
- **Total Thermal Energy:**
$$E_{th} = 600,000 \text{ g} \times 1 \text{ MWd/g} = 600,000 \text{ MWd} = \mathbf{600} \text{ GWd}$$

Note: The breeder extracts $100\times$ *more energy from the same rock.*

(c) The US Depleted Uranium Stockpile

- **Mass:** $700,000$ Metric Tons $= 700,000 \times 1,000$ kg $= 7 \times 10^8$ kg.
- **Total Thermal Energy (using FBR utilization):**
$$E_{th} = (7 \times 10^8 \text{ kg}) \times 0.60 \times (1 \text{ MWd} \cdot 1000 \text{ g/kg}) = 4.2 \times 10^{11} \text{ MWd}$$
- **Electric Energy (40% efficiency):**
$$E_{el} = 0.40 \times 4.2 \times 10^{11} \text{ MWd} = 1.68 \times 10^{11} \text{ MWe-days}$$
- **Convert to TWh:**
$$1 \text{ MWe-day} = 24 \text{ MWh}$$
$$E_{el} = 1.68 \times 10^{11} \times 24 \text{ MWh} = 4.032 \times 10^{12} \text{ MWh}$$
$$E_{el} = \mathbf{4,032,000} \text{ TWh}$$

(d) Years of Power

- **US Consumption:** $\approx 4,000$ TWh/year.

$$\text{Years} = \frac{4,032,000 \text{ TWh}}{4,000 \text{ TWh/yr}} \approx \mathbf{1,000} \text{ years}$$

Conclusion: The "waste" currently sitting in cylinders in Kentucky and Ohio could power the entire US for a millennium carbon-free, if we built breeder reactors.

Problem 2: Burnup and Cycle Length

(a) Reactor Thermal Power

$$P_{th} = \frac{P_{el}}{\eta} = \frac{1200 \text{ MWe}}{0.33} = \mathbf{3,636} \text{ MW}_{\mathbf{th}}$$

(b) Cycle Burnup

- **Energy Produced in one cycle:**

$$E_{cycle} = P_{th} \times t_{days} \times CF$$

$$E_{cycle} = 3636 \text{ MW} \times 540 \text{ days} \times 0.95 = 1,865,268 \text{ MWd}$$

- **Specific Burnup (per mass):**

$$BU_{cycle} = \frac{1,865,268 \text{ MWd}}{90 \text{ MTU}} = \mathbf{20,725} \text{ MWd/MTU}$$

(c) Batches and Limits

- The discharge burnup of a fuel assembly that stays in for N cycles is roughly:

$$BU_{discharge} \approx N \times BU_{cycle}$$

- **For a 3-batch core ($N = 3$):**

$$BU_{discharge} = 3 \times 20,725 = \mathbf{62,175} \text{ MWd/MTU}$$

- **Conclusion:** This **exceeds** the regulatory limit of 62,000 MWd/MTU.
- *Feasibility:* They cannot use a standard 3-batch scheme. They would need to increase the batch size (e.g., replace 40-45% of the core instead of 33%) to lower the residence time, or slightly shorten the cycle length.

Problem 3: Radiation Damage (dpa)

(a) Reaction Rate

- $\sigma_s = 3$ barns $= 3 \times 10^{-24}$ cm^2.
- $\phi = 10^{11}$ n/cm$^2 \cdot$ s.

$$R = \sigma_s \phi = (3 \times 10^{-24}) \times (10^{11}) = \mathbf{3 \times 10^{-13}} \text{ collisions/atom/s}$$

(b) Displacement Rate

$$\text{Rate}_{dpa} = R \times \nu = (3 \times 10^{-13}) \times 500 = \mathbf{1.5 \times 10^{-10}} \text{ dpa/s}$$

(c) Total dpa (60 Years)

- Time $t = 1.9 \times 10^9$ s.

$$\text{Total dpa} = (1.5 \times 10^{-10} \text{ dpa/s}) \times (1.9 \times 10^9 \text{ s}) = \mathbf{0.285} \text{ dpa}$$

(d) Physical Interpretation A value of ≈ 0.3 dpa means that, statistically, **30% of the atoms in the steel vessel have been knocked out of their lattice sites** at least once during the reactor's life. While most snap back into place (annealing), enough defects remain to cause embrittlement.

Problem 4: The DBTT Shift and PTS

Given: Fluence rate $\dot{\Phi} = 2 \times 10^{19}$ n/cm^2/year. Formula: $\Delta\text{DBTT} = 40 + 10(\Phi t/10^{19})^{0.5}$.

(a) 40 Years Operation

- **Total Fluence:** $\Phi_{40} = 40 \text{ yr} \times 2 \times 10^{19} = 80 \times 10^{19}$ n/cm^2.
- **Normalized Fluence Term:** $X = \frac{80 \times 10^{19}}{10^{19}} = 80$.
- **Shift:**
$$\Delta\text{DBTT} = 40 + 10\sqrt{80} = 40 + 10(8.94) = 40 + 89.4 = 129.4^\circ\text{C}$$
- **New DBTT:**
$$T_{new} = T_{initial} + \Delta T = -20^\circ\text{C} + 129.4^\circ\text{C} = \mathbf{109.4}^\circ\text{C}$$

(b) 80 Years Operation

- **Total Fluence:** $\Phi_{80} = 160 \times 10^{19}$ n/cm^2. (Normalized term = 160).
- **Shift:**
$$\Delta\text{DBTT} = 40 + 10\sqrt{160} = 40 + 10(12.65) = 40 + 126.5 = 166.5^\circ\text{C}$$
- **New DBTT:**
$$T_{new} = -20^\circ\text{C} + 166.5^\circ\text{C} = \mathbf{146.5}^\circ\text{C}$$

(c) Safety Check (PTS)

- **Scenario:** ECCS water injects at $T_{water} = 25°\text{C}$.
- **Comparison:** $T_{water}(25°\text{C}) \ll \text{DBTT}_{80yr}(146.5°\text{C})$.
- **Conclusion: YES, the vessel is at risk.** The injection water temperature is far below the transition temperature, meaning the steel will behave in a **brittle** manner (like glass) during the thermal shock. A crack could propagate catastrophically.

Solution to Problem 5: Fuel Rod Mechanics & Ideal Gas Law

Given Data: P_{out} (coolant) = 15.5 MPa. $r = 4.75$ mm. $t = 0.6$ mm.

1. **Scenario A (Unpressurized)**
   - $P_{in} = 0.1$ MPa.
   - $\Delta P = P_{out} - P_{in} = 15.5 - 0.1 = 15.4$ MPa (Net external pressure).
   - **Stress:**
   $$\sigma_\theta = \frac{\Delta P \cdot r}{t} = \frac{15.4 \times 4.75}{0.6} = \mathbf{121.9} \text{ MPa}$$
   - **State:** The cladding is in **Compression** because the external coolant pressure is vastly greater than the internal pressure, pushing the walls inward.

2. **Scenario B (Pre-pressurized, Cold)**
   - $P_{in} = 3.0$ MPa at $T = 300$ K.
   - $\Delta P = P_{out} - P_{in} = 15.5 - 3.0 = 12.5$ MPa.
   - **Stress:**
   $$\sigma_\theta = \frac{\Delta P \cdot r}{t} = \frac{12.5 \times 4.75}{0.6} = \mathbf{99.0} \text{ MPa}$$
   - **State:** Still in **Compression**, but pre-pressurization reduces the stress by about 20% compared to Scenario A.

3. **Scenario C (Pre-pressurized, Hot)**
   - Since the volume (V) and number of moles (n) are constant at the start of life, the Ideal Gas Law simplifies to: $\frac{P_1}{T_1} = \frac{P_2}{T_2}$
   - **Hot Pressure:**
   $$P_{hot} = P_{cold} \times \frac{T_{hot}}{T_{cold}} = 3.0 \times \frac{700}{300} = \mathbf{7.0} \text{ MPa}$$
   - $\Delta P = P_{out} - P_{in} = 15.5 - 7.0 = 8.5$ MPa.
   - **Operating Stress:**
   $$\sigma_\theta = \frac{\Delta P \cdot r}{t} = \frac{8.5 \times 4.75}{0.6} = \mathbf{67.3} \text{ MPa}$$
   - **Improvement:** The magnitude of the stress dropped from 121.9 MPa (Scenario A) to 67.3 MPa. This is a massive **reduction of 54.6 MPa** (∼45% improvement). Heating the gas pushes back against the coolant pressure, drastically reducing the compressive load.

4. **Scenario D (End of Life - EOL)**
   - The number of moles has tripled: $n_{EOL} = 3 \times n_{initial}$.
   - From $P = \frac{nRT}{V}$, if n triples while T and V remain constant, the pressure triples compared to the hot start-of-life state.
   - **New Internal Pressure:** $P_{EOL} = 3 \times 7.0 = \mathbf{21.0}$ MPa.
   - $\Delta P = P_{in} - P_{out} = 21.0 - 15.5 = 5.5$ MPa (Net internal pressure).
   - **Final Stress:**
   $$\sigma_\theta = \frac{\Delta P \cdot r}{t} = \frac{5.5 \times 4.75}{0.6} = \mathbf{43.5} \text{ MPa}$$

5. **Tension & The Limiting Factor**
   - The EOL stress places the cladding in **Tension** because the internal pressure (21.0 MPa) now exceeds the external coolant pressure (15.5 MPa), pushing the walls outward.
   - **Limiting Factor:** If the internal pressure forces the cladding to balloon outward in tension, it pulls away from the fuel pellet. This widens the gap, drastically reducing thermal conductivity. This creates a dangerous positive feedback loop: poorer heat transfer $\rightarrow$ hotter fuel $\rightarrow$ hotter gas $\rightarrow$ higher pressure $\rightarrow$ more ballooning. This ultimately limits how long the fuel can safely remain in the reactor.

6. **Why Helium?**
   - Helium is chemically inert and will not react with the Zircaloy or the UO_2.
   - Most importantly, Helium has an exceptionally high **Thermal Conductivity** compared to heavier gases like Argon or Nitrogen. This ensures efficient heat transfer across the gap from the fuel pellet to the cladding, keeping the fuel centerline temperatures safely below melting.